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United States Patent	12415636
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Song; Jianyu et al.

Takeoff and landing platform, unmanned aerial vehicle, takeoff and landing system, storage device and takeoff and landing control method

Abstract

A takeoff and landing platform, a UAV, a takeoff and landing system, a storage device and a takeoff and landing control method are provided. The takeoff and landing platform includes: a bracket, one end of the bracket is fixed on a base, and another end thereof extends in a direction away from the base, and the bracket is provided with a vertical guide rail. Multiple UAVs may be vertically stacked on the bracket along the guide rail and take off from the bracket. Hence, the manpower investment and site investment are reduced for multiple UAVs to perform collaborative operations. It can not only reduce the cost of multiple UAV collaborative operations, but also improve the efficiency of multiple UAV collaborative operations.

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Appl. No.: **18/395107**

Filed: **December 22, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240124169 A1	Apr. 18, 2024

Related U.S. Application Data

continuation parent-doc WO PCT/CN2021/103660 20210630 PENDING child-doc US 18395107

Publication Classification

Int. Cl.: **B64U70/97** (20230101); **B60L53/14** (20190101); **B64U70/92** (20230101); **B64U80/25** (20230101); **B64U80/40** (20230101)

U.S. Cl.:

CPC **B64U70/97** (20230101); **B60L53/14** (20190201); **B64U70/92** (20230101); **B64U80/25** (20230101); **B64U80/40** (20230101); B60L2200/10 (20130101)

Field of Classification Search

CPC: B64U (70/97); B64U (70/92); B64U (80/25); B64U (80/40); B64U (50/37); B60L (53/14); B60L (2200/10); B64C (39/024); B64C (5/04); B64F (1/02); Y02T (10/70); Y02T (10/7072); Y02T (90/14)

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation application of PCT application No. PCT/CN2021/103660, filed on Jun. 30, 2021, and the content of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD
(2) The present disclosure relates to the technical field of unmanned aerial vehicles (UAVs), and in particular to a takeoff and landing platform for multiple UAVs, a UAV, a UAV takeoff and landing system, a storage device and a UAV takeoff and landing control method.
BACKGROUND
(3) With the rapid development in the UAV technologies, the functions of UAVs are becoming more and more diversified, and the application scenarios of UAVs are becoming more and more extensive. Correspondingly, there are more and more scenarios that require multiple UAVs to perform collaborative operations. For example, in scenarios such as UAV performance or UAV transportation of emergency supplies, multiple UAVs are often needed to work together.
(4) In the existing technologies, in the cases where multiple UAVs need to be used for collaborative operations, during the stages of UAV takeoff site layout, UAV preparation and battery replacement, and UAV landing and recovery, etc., the operations need high manpower investments and large sites. This may not only increase the cost of the collaborative operation of multiple UAVs, but also reduce the efficiency of the collaborative operation.
SUMMARY
(5) In order to solve the problems of high cost and low efficiency of multiple UAV cooperative operations in the existing technology, embodiments of the present disclosure provide a takeoff and landing platform for multiple UAVs, a UAV, a UAV takeoff and landing (docking) system, a storage device, and a UAV takeoff and landing control method.
(6) In one aspect, exemplary embodiments of the present disclosure provide a platform, including a base; a bracket, including a guide rail extended upright from the base; an accommodating portion defined in the bracket and extended along the guide rail, the accommodating portion including an opening away from the base, where the accommodating portion is configured to accommodate the plurality of UAVs which are stacked on the bracket along the guide rail and take off from the opening.
(7) In another aspect, exemplary embodiments of the present disclosure provide a docking system, including a platform including: a bracket, including a vertical guide rail; and a plurality of unmanned aerial vehicles (UAVs), where each UAV includes: one of a protruding structure and a recessing structure disposed at a bottom part of the UAV, the other one of the protruding structure and the recessing structure disposed on a top part of the UAV, where the protruding structure is configured to be at least partially embedded in the recessing structure to facilitate vertical stacking of the UAV, where the plurality of UAVs is configured to be stacked vertically on the takeoff and landing platform, the plurality of UAVs is configured to be vertically stacked on the takeoff and

landing platform in a landing process, and the plurality of UAVs is configured to take off from the takeoff and landing platform in a takeoff process.
(8) In the embodiments of the present application, by providing a bracket(s) on the takeoff and landing platform, when multiple UAVs land, the multiple UAVs may be vertically stacked on the bracket along a guide rail; when multiple UAVs take off, the multiple UAVs may take off from the bracket. That is to say, the takeoff and landing platform may be used to realize the takeoff and landing of multiple UAVs. In this way, it is possible to avoid manual UAV takeoff site layout, preparation, and landing recovery operations, and reduce manpower and site investment when multiple UAVs perform collaborative operations. It may not only reduce the cost of multiple UAV collaborative operations, but also improve the efficiency of multiple UAV collaborative operations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In order to more clearly illustrate the technical solutions of the embodiments of the present disclosure, the following will briefly introduce the drawings for the description of some exemplary embodiments. Apparently, the accompanying drawings in the following description are some exemplary embodiments of the present disclosure. For a person of ordinary skill in the art, other drawings may also be obtained based on these drawings without creative efforts.
(2) FIG. **1** is a schematic diagram of the structure of a takeoff and landing platform according to some exemplary embodiments of the present disclosure;
(3) FIG. **2** is a schematic diagram of the structure of a takeoff and landing system using the takeoff and landing platform shown in FIG. **1**;
(4) FIG. **3** is a schematic diagram of the structure of the takeoff and landing system shown in FIG. **2** viewed from another angle;
(5) FIG. **4** is a schematic diagram of the structure of the takeoff and landing system shown in FIG. **2** viewed from another angle;
(6) FIG. **5** is a schematic diagram of the structure of a first bracket according to some exemplary embodiments of the present disclosure;
(7) FIG. **6** is a schematic exploded diagram of the structure of the first bracket shown in FIG. **5**;
(8) FIG. **7** is a schematic diagram of the structure of a second bracket according to some exemplary embodiments of the present disclosure;
(9) FIG. **8** is a schematic exploded diagram of the structure of the second bracket shown in FIG. **7**;
(10) FIG. **9** is a schematic diagram of the structure of a guide carrier plate **12** according to some exemplary embodiments of the present disclosure;
(11) FIG. **10** is a schematic diagram of the structure of a takeoff and landing platform according to some exemplary embodiments of the present disclosure;
(12) FIG. **11** is a schematic diagram of the structure of a takeoff and landing platform according to some exemplary embodiments of the present disclosure;
(13) FIG. **12** is a schematic diagram of the structure of the takeoff and landing platform shown in FIG. **11** viewed from another angle;
(14) FIG. **13** is a schematic diagram of the structure of a takeoff and landing platform according to some exemplary embodiments of the present disclosure;
(15) FIG. **14** is a schematic diagram of the structure of a takeoff and landing platform shown in FIG. **13**;
(16) FIG. **15** is a schematic diagram of the structure of a UAV according to some exemplary embodiments of the present disclosure;
(17) FIG. **16** is a schematic diagram of the structure of the UAV shown in FIG. **15** viewed from another angle;

(18) FIG. **17** is a schematic diagram of the structure of an A-A section of the UAV shown in FIG. **15**;
(19) FIG. **18** is a schematic diagram of the structure of a storage device according to some exemplary embodiments of the present disclosure;
(20) FIG. **19** is a schematic diagram of the structure of the storage device shown in FIG. **18** viewed from another angle;
(21) FIG. **20** is a schematic diagram of the structure of the storage device shown in FIG. **18** viewed from another angle;
(22) FIG. **21** is a schematic diagram of the structure of a storage device according to some exemplary embodiments of the present disclosure;
(23) FIG. **22** is a schematic diagram of the structure of the storage device shown in FIG. **21** viewed from another angle;
(24) FIG. **23** is a schematic diagram of the structure of a first telescopic frame according to some exemplary embodiments of the present disclosure;
(25) FIG. **24** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure;
(26) FIG. **25** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure;
(27) FIG. **26** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure;
(28) FIG. **27** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure; and
(29) FIG. **28** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure.
(30) Description of element symbols in the drawings: **10**—bracket, **101**—guide rail, **102**—first bracket, **1021**—first inclined surface, **103**—second bracket, **1031**—second inclined surface, **104**—casing, **105**—sealing plate, **106**—opening, **11**—first charging module, **12**—guide carrier plate, **121**—lower recess part, **122**—first guide part, **123**—first opening, **124**—second opening, **13**—lifting module, **131**—driving member, **132**—screw, **133**—sliding block, **134**—support block, **14**—flow guide frame, **141**—flow guide opening, **142**—flow guide plate, **143**—support frame, **15**—lifting device, **151**—lifting mechanism, **152**—accommodating bin, **153**—connecting mechanism, **16**—base, **17**—auxiliary moving device, **18**—air flow generating device, **181**—air flow outlet, **100**—UAV, **20**—body, **201**—bearing platform, **21**—arm, **22**—recessing structure, **23**—protruding structure, **231**—planar part, **232**—second guide part, **24**—connecting frame, **241**—third opening, **242**—fourth opening, **25**—propeller, **26**—propeller protective cover, **30**—storage platform, **301**—first telescopic frame, **3011**—first frame body, **3012**—first telescopic mechanism, **302**—second telescopic frame, **31**—takeoff and landing platform.

DETAILED DESCRIPTION

(31) The technical solutions in some exemplary embodiments of the present disclosure will be described below with reference to the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are part of the embodiments of the present disclosure, but not all of the embodiments. Based on the exemplary embodiments in the present disclosure, all other embodiments obtained by a person of ordinary skill in the art without creative efforts fall within the scope of protection of the present disclosure.
(32) The features with “first” and “second” in the description and claims of this disclosure may include one or more of these features, either explicitly or implicitly. In the description of this disclosure, unless otherwise stated, “a plurality of” means two or more. In addition, “and/or” in the description and claims indicates at least one of the related objects, and the character “/” generally indicates that the related objects are in an “or” relationship.
(33) In the description of this disclosure, it needs to be understood that the orientation or positional

relationships indicated by terms like “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “back”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inside”, “outside”, “clockwise”, “counterclockwise”, “axial”, “radial”, “circumferential”, etc. are based on the orientation or positional relationship shown in the drawings. They are merely to facilitate the description of the present disclosure and to simplify the description, and are not intended to indicate or imply that the device or element referred to must have a specific orientation, or be constructed and operate in a specific orientation. Therefore, they cannot be construed as a limitation on this disclosure.
(34) In the description of the present disclosure, it should be noted that, unless otherwise explicitly stipulated and limited, the terms “mounting”, “connecting” and “linking” should be understood in a general sense. For example, it may be a fixed connection, a detachable connection, or an integral connection; can be a mechanical connection or an electrical connection; can be a direct connection, or an indirect connection via an intermediate medium, or can be an internal connection between two components. For a person of ordinary skill in the art, the specific meanings of the above terms in the present disclosure can be understood on a case-by-case basis.
(35) Reference may be made to the following figures, FIG. **1** is a schematic diagram of the structure of a takeoff and landing platform according to some exemplary embodiments of the present disclosure; FIG. **2** is a schematic diagram of the structure of a takeoff and landing system using the takeoff and landing platform shown in FIG. **1**; FIG. **3** is a schematic diagram of the structure of the takeoff and landing system shown in FIG. **2** viewed from another angle; FIG. **4** is a schematic diagram of the structure of the takeoff and landing system shown in FIG. **2** viewed from another angle. The takeoff and landing platform may specifically include:
(36) A bracket **10**, where the bracket **10** is provided with a vertical/upright guide rail(s) **101**;
(37) Multiple UAVs **100** may be vertically stacked on the bracket **10** along the vertical/upright guide rail(s) **101**, and may take off from the bracket **10**.
(38) In the present disclosure, the bracket **10** is provided on the takeoff and landing platform. In the case of multiple UAVs **100** landing, the multiple UAVs **100** may be vertically stacked on the bracket **10** along the guide rail **101**. In the case of the multiple UAVs **100** taking off, the multiple UAVs **100** may take off from the bracket **10**. That is, the takeoff and landing platform may be used to implement the takeoff and landing of the multiple UAVs **100**. In this way, it is possible to avoid manual operations of UAV **100** takeoff site layout, preparation, and landing recovery, and reduce manpower and site investment when multiple UAVs **100** perform collaborative operations. It may not only reduce the cost of multiple UAV **100** collaborative operations, but also improve the efficiency of multiple UAV **100** collaborative operations.
(39) For example, in a UAV performance scenario where multiple UAVs **100** perform collaborative operations, multiple UAVs **100** may be stacked and stored on the takeoff and landing platform. When the multiple UAVs **100** need to take off, the multiple UAVs **100** may take off from the bracket **10** on the takeoff and landing platform. In this way, the manual operation for arranging the takeoff site and placing multiple UAV **100***s* one by one on the takeoff site may be avoided. After the performance is completed, the multiple UAVs **100** may be vertically stacked on the bracket **10** along the guide rail(s) **101** on the bracket **10**, thus avoiding the problem of the UAVs **100** first landing at the takeoff site and then relying on manual labor to recover and store the multiple UAVs **100**. On the one hand, this may reduce the venue investment and manpower investment required for UAV **100** performances; on the other hand, the efficiency of UAV **100** performances may also be improved.
(40) In some exemplary embodiments, in the case where multiple UAVs **100** are stacked vertically on the bracket **10**, the takeoff and landing platform may include a storage state and a charging state; where in the storage state, the takeoff and landing platform is used to store the multiple UAVs **100**; in the charging state, the takeoff and landing platform may be used to charge the multiple UAVs **100** to improve the endurance of the UAVs **100**.

(41) In a specific application, when the takeoff and landing platform is in the storage state, the multiple UAVs **100** on the takeoff and landing platform may be arranged as compactly as possible to reduce the volume occupied by the multiple UAVs **100** and facilitate the reduction of the space required for storage and transportation of the takeoff and landing platform and the UAV **100**. When the takeoff and landing platform is in the charging state, the distance between the UAVs **100** should be reasonably controlled to facilitate charging of multiple UAVs **100** and improve charging safety. Therefore, the positions of the multiple UAVs **100** relative to the bracket **10** in the storage state may be different from the positions of the multiple UAVs **100** relative to the bracket **10** in the charging state.
(42) In some exemplary embodiments, the takeoff and landing platform may also be provided with: a plurality of first charging modules **11** stacked vertically on the bracket **10**. In the charging state, the first charging modules **11** may be electrically connected to a second charging module of the UAV **100** to charge the UAV **100** and improve the endurance of the UAV **100**.
(43) Specifically, the takeoff and landing platform may also be provided with a control module. The control module may be in communication with the plurality of first charging modules **11** to control the first charging modules **11** to charge the UAVs **100**.
(44) In some exemplary embodiments, the first charging module **11** may include a charging interface and a signal transmission interface. The charging interface may be used to charge the UAV **100**; the signal transmission interface may be used to implement data exchange between the first charging module **11** and the second charging module.
(45) Specifically, when the first charging module **11** and the second charging module are interface-connected, the charging interface and the signal transmission interface may be interface-connected to the second charging module at the same time. The charging interface may be electrically connected to a charging interface on the UAV **100** to charge the UAV **100**. The signal transmission interface may be connected to a signal interface on the UAV **100** to exchange data between the first charging module **11** and the UAV **100**, for example, download the data of the UAV **100**, update the firmware of the UAV **100**, identify the code of the UAV **100**, or enter the route, etc.
(46) Specifically, the height of the UAV **100** is the first height, and the vertical interval between two adjacent first charging modules **11** is the first interval, where the first interval is greater than the first height. In this way, it would be easy to align the second charging module on the UAV **100** with the first charging module **11** on the takeoff and landing platform.
(47) In practical applications, the interval between UAVs **100** may be adjusted with reference to the first interval between the first charging modules **11**, so that the second charging module on the UAV **100** may be aligned with the first charging module **11** on the takeoff and landing platform.
(48) In some exemplary embodiments, the bracket **10** may be provided with a storage space for accommodating the multiple UAVs **100**. Specifically, when the multiple UAVs **100** are stacked on the bracket **10**, the multiple UAVs **100** may be stored in the storage space.
(49) In practical applications, the guide rail **101** and the first charging module **11** may both be disposed in the storage space. This is to facilitate the guide rail **101** to guide the stack of the UAV **100**, and to facilitate the first charging module **11** to charge the UAV **100**.
(50) As shown in FIG. **2**, the bracket **10** may include a first bracket **102** and a second bracket **103**, where the first bracket **102** and the second bracket **103** are relatively spaced apart. The storage space is formed between the first bracket **102** and the second bracket **103**. The first bracket **102** and the second bracket **103** are each provided with a guide rail **101** to guide the UAVs **100** from both sides and introduce the UAVs **100** into the storage space, and support the UAVs **100** from both sides to improve the stacking reliability of the UAVs **100**.
(51) In some exemplary embodiments, the guide rails **101** of the first bracket **102** and the second bracket **103** are at least partially embedded in the multiple UAVs **100** to limit the movement of the UAVs **100** in the storage space, such that the UAVs **100** are prevented from shaking in the storage space, so as to improve the stacking reliability of the UAVs **100** in the storage space.

(52) In some exemplary embodiments, the number of the bracket **10** is one, and the storage space is formed on the bracket **10**. Specifically, in order to improve the stacking reliability of the UAVs **100** on the bracket **10**, the storage space may be located at a center of the bracket **10**.
(53) In some exemplary embodiments, the UAV **100** may be provided with a mounting through hole(s). In the case where multiple UAVs **100** are vertically stacked on the bracket **10**, the mounting through holes of the UAVs **100** may be sleeved on the bracket **10**. Specifically, the mounting through hole may be provided in a central area of the UAV **100** to improve the stacking reliability of the UAV **100** on the bracket **10**.
(54) In practical applications, when the UAV **100** needs to land, the mounting through hole on the UAV **100** may be aligned with the bracket **10**. In this way, the UAV **100** is then landed to connect the mounting through holes on the UAV **100** with the bracket **10** to realize stacking of the UAV **100** on the bracket **10**.
(55) In some exemplary embodiments, the guide rails **101** on the bracket **10** may be at least partially embedded in the multiple UAVs **100**, such that the movement of the UAV **100** in the storage space is limited to prevent the UAV **100** from shaking on the bracket **10**.
(56) In specific applications, the UAV **100** may be provided with an opening. When multiple UAVs **100** need to be stacked vertically, the openings on the UAVs **100** may be aligned with the guide rail **101** on the bracket **10**, and then the UAV **100** is controlled to descend, such that the guide rail **101** on the bracket **10** is embedded in the opening of the UAV **100**.
(57) It should be noted that in the drawings of the present disclosure, only the case where the number of brackets **10** is two and the storage space is formed between the two brackets **10** is shown. In practical applications, the number of brackets **10** may be selected according to actual conditions. For example, the number of brackets **10** may be 1, 2, 3 or 5, etc. The present disclosure does not limit the number of the bracket **10**. When the number of the bracket **10** is multiple (two or more), the storage space may be formed between the multiple bracket **10**.
(58) In specific applications, the takeoff and landing platform may also include: a guide carrier plate **12**. The guide carrier plate **12** is at least partially disposed in the storage space. The guide carrier plate **12** is used to carry the UAV **100**.
(59) Specifically, the guide carrier plate **12** may be movably connected to the bracket **10**. During a takeoff process of the UAV **100**, the guide carrier plate **12** may move upward to lift the UAV **100**. During the landing process of the UAV **100**, the guide carrier plate **12** may move downward, so that the UAV **100** may vertically descends and be stacked on the bracket **10**.
(60) FIG. **5** is a schematic diagram of the structure of a first bracket according to some exemplary embodiments of the present disclosure. FIG. **6** is a schematic exploded diagram of the structure of the first bracket shown in FIG. **5**. FIG. **7** is a schematic diagram of the structure of a second bracket according to some exemplary embodiments of the present disclosure. FIG. **8** is a schematic exploded diagram of the structure of the second bracket shown in FIG. **7**.
(61) As shown in FIG. **5** to **8**, the guide carrier plate **12** may be connected to the first bracket **102** and the second bracket **103**, and the first charging module **11** may be connected to the second bracket **103**.
(62) Referring to FIG. **9**, a schematic structural diagram of a guide carrier plate **12** according to some exemplary embodiments of the present disclosure is shown. As shown in FIG. **9**, the guide carrier plate **12** may include a lower recess part **121** and a first guide part **122**, where the lower recess part **121** is arranged in a central area of the guide carrier plate **12**; the first guide part **122** is arranged at an edge area of the guide carrier plate **12** and surrounds the lower recess **121**; one side of the first guide part **122** is connected to an edge of the lower recess part **121**, and another side of the first guide part **122** extends in a direction away from the lower recess part **121** and obliquely upward.
(63) Specifically, the lower recess part **121** may be located at a center of the storage space, and used to connect with a bottom of the body of the UAV **100** and support the UAV **100**. The first

guide part **122** may be used to guide the UAV **100** into the lower recess part **121**.
(64) In some exemplary embodiments, the lower recess part **121** may have a planar structure to fully contact the bottom of the UAV **100**, thereby improving the support reliability of the guide carrier plate **12** for the UAV **100**.
(65) In some exemplary embodiments, the first guide part **122** may have an inclined planar structure or a curved surface structure. In this way, during the landing of the UAV **100**, the UAV **100** may slide along the inclined first guide part **122** into the lower recess part **121** to achieve reliable support of the UAV **100** by the guide carrier plate **12**.
(66) In some exemplary embodiments, in the case where the bracket **10** includes a first bracket **102** and a second bracket **103**, the first guide part **122** may be provided with a first opening **123** and a second opening **124**. The first opening **123** may be sleeved over the guide rail **101** of the first bracket **102**, and the second opening **124** may be sleeved over the guide rail **101** of the second bracket **103**, so as to limit the movement of the guide carrier plate **12** on the first bracket **102** and the second bracket **103**.
(67) In specific applications, the first opening **123** and the second opening **124** may be arranged on the first guide part **122** of the guide carrier plate **12**, the first opening **123** is sleeved over the guide rail **101** on the first bracket **102**, and the second opening **124** is sleeved over the guide rail **101** on the second bracket **103**. Thus, this may limit the guide carrier plate **12** to only slide along the guide rail **101** on the first bracket **102** and the second bracket **103**, thereby preventing the guide carrier plate **12** from shaking on the first bracket **102** and the second bracket **103**.
(68) FIG. **10** is a schematic diagram of the structure of a takeoff and landing platform according to some exemplary embodiments of the present disclosure. As shown in FIG. **10**, a top part of the guide rail **101** of the first bracket **102** is provided with a first inclined surface **1021**, and a top part of the guide rail **101** of the second bracket **103** is provided with a second inclined surface **1031**. The shape of the first inclined surface **1021** and the second inclined surface **1031** may be adapted to the shape of the first guide part **122**. When the guide carrier plate **12** moves to the top of the guide rail **101**, the first inclined surface **1021** and the second inclined surface **1031** may be flush with the first guide part **122** to form a complete profile to better guide the UAV **100** into the lower recess part **121** in the central area of the guide carrier plate **12**.
(69) Specifically, the shape of the first inclined surface **1021** and the second inclined surface **1031** may be adapted to the first guide part **122** as follows: the cross-sectional shape of the first inclined surface **1021** and the second inclined surface **1031** may be the same as the cross-sectional shape of the first guide part **122**. For example, when the cross-sectional shape of the first guide part **122** is an inclined plane, the cross-sectional shapes of the first inclined surface **1021** and the second inclined surface **1031** may be correspondingly inclined planes. In another example, when the cross-sectional shape of the first guide part **122** is an inclined curved surface, the cross-sectional shapes of the first inclined surface **1021** and the second inclined surface **1031** may be respectively inclined curved surfaces.
(70) In some exemplary embodiments, the bracket **10** is also provided with a lifting module **13**. The lifting module **13** may slide along the bracket **10** and is connected to the guide carrier plate **12**. The lifting module **13** may be used to lift the guide carrier plate **12**. In practical applications, by sliding the lifting module **13** along the bracket **10**, the guide carrier plate **12** may be driven to slide up and down along the bracket **10**, so as to lift the UAV **100** on the guide carrier plate **12** to the top of the bracket **10**, or lower the UAV **100** on the guide carrier plate **12** to the bottom of the bracket **10**.
(71) In some exemplary embodiments, the lifting module **13** may include: a driving member **131**, a screw **132** and a sliding block **133**, where an output end of the driving member **131** may be connected to the screw **132** for driving the screw **132** to rotate, and the axial direction of the screw **132** is parallel to the guide direction of the guide rail **101**; the sliding block **133** is provided with a threaded hole, the threaded hole is sleeved over the screw **132** and connected to the thread of the screw **132**; the rotation of the screw **132** may drive the sliding block **133** to slide along the axial

direction of the screw **132**. The sliding block **133** may be used to connect to the guide carrier plate **12** to drive the guide carrier plate **12** to slide along the bracket **10**.
(72) Specifically, the driving member **131** may be a servo motor. The servo motor may be connected to a bottom part of the screw **132**. An output end of the servo motor may be connected to the screw **132**. When the servo motor rotates, the screw **132** may be driven to rotate accordingly. Moreover, the rotation direction of the screw **132** may be consistent with the rotation direction of the servo motor.
(73) It should be noted that in practical applications, in order to achieve reliable lifting of the guide carrier plate **12**, the lifting module **13** may be provided for both the first bracket **102** and the second bracket **103**.
(74) In some exemplary embodiments, the bracket **10** may also include: a casing **104** and a sealing plate **105** that can be matched and connected to each other. A accommodation cavity may be formed between the casing **104** and the sealing plate **105**, where the guide rail **101** is provided on the casing **104**; the driving member **131** and the screw **132** are both arranged in the accommodation cavity. At least part of the sliding block **133** is exposed and extends outside the guide rail **101** to connect to the guide carrier plate **12**.
(75) As shown in FIG. **6**, the first bracket **102** may be enclosed by the casing **104** and the sealing plate **105**. As shown in FIG. **8**, the second bracket **103** may also be enclosed by the casing **104** and the sealing plate **105**.
(76) In specific applications, by arranging the driving member **131** and the screw **132** in the accommodation cavity formed by the casing **104** and the sealing plate **105**, this may achieve a protective effect on the driving member **131** and the screw **132** to prevent external water and impurities from entering the interior of the driving member **131** and the screw **132** to cause a short circuit of the driving member **131** or affect the movement accuracy of the screw **132**. Thus, it may improve the working reliability of the lifting module **13**.
(77) Specifically, since the sliding block **133** at least partially extends outside the guide rail **101** and is configured to slide up and down along the guide rail **101**, a long opening **106** may be provided on the guide rail **101** to facilitate the sliding of the slider **133** along the opening **106**. The extending direction of the opening **106** may be consistent with the extending direction of the guide rail **101**.
(78) In some exemplary embodiments, the lifting module **13** may also include a support block **134**. One end of the support block **134** may be fixed on the sliding block **133**, and another end thereof may extend outside the guide rail **101** and be connected to a bottom part of the guide carrier plate **12** to fully support the guide carrier plate **12** and improve the support reliability from the lifting module **13** for the guide carrier plate **12**.
(79) In some exemplary embodiments, the takeoff and landing platform may also include a control module. The control module is electrically connected to the lifting module **13**. When multiple UAVs **100** land, the control module may control the lifting module **13** to drive the guide carrier plate **12** to descend, so that the multiple UAVs **100** may be vertically stacked on the bracket **10** along the guide rail **101**. When the multiple UAVs **100** take off, the control module may control the lifting module **13** to drive the guide carrier plate **12** to rise, so that the multiple UAVs **100** may take off based on the bracket **10**.
(80) Specifically, the control module may be electrically connected to the driving member **131** in the lifting module **13** to control the driving member **131** to rotate forward or reverse so as to drive the guide carrier plate **12** to rise or fall; thus the UAV **100** is driven to rise or fall by the rise or fall of the guide carrier plate **12**.
(81) For example, the control module may be disposed in the accommodation cavity inside the bracket **10**, or may be disposed in the base **16** at the bottom of the bracket **10**. The present disclosure does not limit the location of the control module.
(82) In some exemplary embodiments, the takeoff and landing platform may also include: an elastic receiving part. At least part of the elastic receiving member may be arranged in the storage

space. The elastic receiving member may be used to receive the UAV **100**.

(83) In practical applications, during the landing of the UAV **100**, the UAV **100** may first land on the elastic receiving member. Since the elastic receiving member is at least partially disposed in the storage space, the elastic compression of the elastic receiving member may not only buffer the impact of the falling UAV **100**, but also guide the UAV **100** into the storage space.

(84) In some exemplary embodiments, the compression direction of the elastic receiving member may be consistent with the extension direction of the guide rail **101**. In this way, during the landing of the UAV **100**, with the compression of the elastic receiving member along the compression direction thereof, the UAV **100** may be driven to slide along the guide rail **101**, which may improve the stacking efficiency of multiple UAVs **100**.

(85) In some exemplary embodiments, the elastic receiving member may include at least one of a spring or an elastic piece. The present disclosure does not specifically limit the specific type of the elastic receiving member.

(86) In some exemplary embodiments, the takeoff and landing platform may also include: a guide member. The guide member is connected to the bracket **10**. The guide member may be used to guide the UAV **100** into the storage space, so that the UAV **100** may be quickly stacked in the storage space of the bracket **10**.

(87) Specifically, the guide member may include at least one of a mechanical guide member, an electric guide member, or a pneumatic guide member. The present disclosure does not limit the specific type of the guide member.

(88) FIG. **11** is a schematic diagram of the structure of a takeoff and landing platform according to some exemplary embodiments of the present disclosure. FIG. **12** is a schematic diagram of the structure of the takeoff and landing platform shown in FIG. **11** viewed from another angle. As shown in FIGS. **11** and **12**, the guide member may include a flow guide frame **14**. The flow guide frame **14** may be arranged on the top part of the bracket **10** and surrounds the storage space. The flow guide frame **14** may utilize the downward airflow generated by the UAV **100** to push the UAV **100** toward the storage space.

(89) Specifically, the flow guide frame **14** may be arranged around the storage space of the bracket **10**. During the landing process of the UAV **100**, the UAV **100** may generate downward airflow. When the downward airflow acts on the flow guide frame **14**, the downward airflow may rebound due to the blocking effect of the flow guide frame **14**. Thus, the rebound airflow acts on the UAV **100**, the UAV **100** may be pushed toward the storage space.

(90) In some exemplary embodiments, the flow guide frame **14** may be provided with a flow guide opening **141**. The flow guide opening **141** is opposite to the storage space. Specifically, the flow guide frame **14** may push the UAV **100** to the flow guide opening **141**, so that the UAV **100** may enter the storage space opposite to the flow guide opening **141** to achieve the stacking of the UAVs **100** in the storage space.

(91) In some exemplary embodiments, the flow guide frame **14** may include a plurality of flow guide plates **142**. The plurality of flow guide plates **142** may be connected in sequence around the circumference of the flow guide opening **141**. One side of the flow guide plate **142** is close to the flow guide opening **141**, and another side thereof extends in an upward direction away from the flow guide opening **141**. In this way, by tilting the flow guide plates **142** around the flow guide opening **141**, the UAV **100** may be pushed towards the flow guide opening **141**.

(92) Specifically, the plurality of flow guide plates **142** may be in a circular shape and connected around the circumferential direction of the flow guide opening **141**, or may be in a rectangular, pentagonal or hexagonal shape and connected around the circumferential direction of the flow guide opening **141**. The present disclosure does not specifically limit the surrounding shape of the plurality of flow guide plates **142**. In practical applications, two adjacent flow guide plates **142** may be fixedly connected by snap-on connection, fastener connection, or other connection methods. The present disclosure may not specifically limit the fixed connection method between the flow guide

plates **142**.
(93) In some exemplary embodiments, the flow guide plate **142** may be an inclined flat plate or arc-shaped plate, and the present disclosure does not limit the specific shape of the flow guide plate **142**.
(94) In some exemplary embodiments, the flow guide frame **14** may also include: a support frame **143**, which is connected to the flow guide plate **142** to support the flow guide plate **142**. The support frame **143** may be provided around the bracket **10**.
(95) In practical applications, the support frame **143** may be fixed, together with the bracket **10** on the base **16** at the bottom of the bracket **10**, or may be arranged separately from the bracket **10**. The present disclosure does not specifically limit the connection method between the support frame **143** and the bracket **10**.
(96) In some exemplary embodiments, the flow guide frame **14** is also equipped with a buffer pad. The buffer pad may be used to buffer the impact force when the UAV **100** landing on the flow guide frame **14**.
(97) Specifically, during the landing process of the UAV **100**, the UAV **100** may fail to accurately land on the guide opening **141** but land on the guide plate **142**. By arranging a buffer pad on the flow guide plate **142** of the flow guide frame **14**, the impact when the UAV **100** comes into contact with the flow guide plate **142** may be reduced, and the service life of the flow guide frame **14** may be improved.
(98) In some exemplary embodiments, the buffer pad may include: at least one of a rubber pad or a foam pad. The present disclosure does not limit the specific type of the buffer pad.
(99) In some exemplary embodiments, the guide member may include: an air flow generating device **18**; the air flow generating device **18** is arranged on the top of the first bracket **102** and the second bracket **103**, and is arranged around the storage space, a plurality of airflow devices **18** may push the UAV **100** toward the storage space.
(100) Specifically, the air flow generating device **18** can generate an air flow. Since the air flow generating devices **18** is arranged around the storage space, when UAV **100** is landing, the air flow generated by the air flow generating device **18** may push the UAV **100** toward the storage space, thereby facilitating the stacking of the UAV **100** in the storage space.
(101) In some exemplary embodiments, the air flow generating device **18** may include: a plurality of air flow outlets **181**. Each air flow outlet **181** may release an air flow to propel the UAV **100**. The plurality of air flow outlets **181** may be positioned around the storage space to push the UAV **100** toward the storage space from multiple directions. In this way, even if the landing position of the UAV **100** is relatively complex, through the synergy of the plurality of air flow outlets, the UAV **100** may still be accurately pushed to the storage space, thereby improving the guidance accuracy of the air flow generating device **18** for the UAV **100**.
(102) It should be noted that in practical applications, multiple air flow outlets **181** may share one air flow generating device **18**, or each air flow outlet may correspond to an independent air flow generating device **18**. The present disclosure does not limit this.
(103) FIG. **13** is a schematic diagram of the structure of a takeoff and landing platform according to some exemplary embodiments of the present disclosure; FIG. **14** is a schematic diagram of the structure of a takeoff and landing platform shown in FIG. **13**. As shown in FIGS. **13** and **14**, the takeoff and landing platform may also include: a lifting device **15**. The lifting direction of the lifting device **15** is consistent with the extending direction of the bracket **10**, where when the UAV **100** is stacked on the bracket **10**, the lifting device **15** may transport a target object (target) carried on the UAV **100**.
(104) Specifically, the lifting device **15** may be used in conjunction with the bracket **10**. The lifting device **15** is consistent with the extending direction of the bracket **10** and may be raised or lowered in the vertical direction. When multiple UAVs **100** are stacked on the bracket **10**, the lifting device **15** may transport the target object carried on the UAV **100** to a location other than the bracket **10**.

Alternatively, the target object is transported to the UAV **100** from an external location other than the bracket **10** to realize the input and output of the target object.
(105) In some exemplary embodiments, the target may be passengers or cargo. For example, in the case where the UAV **100** is a passenger UAV **100**, the target may be a passenger. In another example, in the case where the UAV **100** is a logistics UAV **100**, the target object may be cargo. The present disclosure does not limit the specific content of the target object.
(106) In some exemplary embodiments, the lifting device **15** may specifically include a lifting mechanism **151** and an accommodating bin **152**. The accommodating bin **152** may be used to accommodate the target object; and the accommodating bin **152** is connected to the lifting mechanism **151**. The lifting mechanism **151** may drive the accommodating bin **152** to rise or fall to transport the target object.
(107) Specifically, the lifting mechanism **151** may be provided with a lifting bracket and a lifting driving mechanism. The lifting driving mechanism may be connected to the accommodation bin **152** to drive the accommodation bin **152** to rise or fall along the lifting bracket.
(108) For example, the lifting driving mechanism may include a motor, a push rod, or other driving mechanism that may drive the accommodation bin **152** to rise or fall. The present disclosure does not specifically limit the lifting driving mechanism.
(109) Specifically, the accommodation bin **152** may include a cabin body and a bin door for accommodating the target object. When the bin gate is open, it is convenient for the target object to enter and exit the cabin; when the cabin door is closed, the target object in the cabin may be protected.
(110) As shown in FIG. **14**, the lifting device **15** may also include: a connecting mechanism **153**. The connecting mechanism **153** is connected to the accommodation bin **152**. In the case where the accommodation bin **152** is aligned with a target UAV, the connecting mechanism **153** may be connected between the accommodation bin **152** and the target UAV, so as to transfer the target object from the target UAV to the accommodation bin **152**, or transfer the target object from the accommodation bin **152** to the target UAV.
(111) Specifically, the target UAV may be a UAV **100** among the multiple UAVs **100** for transporting the target object. For example, when the multiple UAVs **100** are stacked vertically on the bracket **10**, if the top UAV **100** needs to transport the target object, the top UAV **100** may be used as the target UAV.
(112) In some exemplary embodiments, the connecting mechanism **153** serves as a bridge connecting the accommodation bin **152** and the target UAV. By means of the connecting mechanism **153**, the target object in the accommodation bin **152** can be transferred to the target UAV; alternatively, the target object in the target UAV can be transferred to the accommodation bin **152**.
(113) In some exemplary embodiments, the connecting mechanism **153** may specifically include an unfolded state and a folded state. As shown in FIG. **14**, in the unfolded state, the connecting mechanism **153** may connect between the accommodation bin **152** and the target UAV to realize the transfer of the target object between the accommodation bin **152** and the target UAV. As shown in FIG. **13**, in the folded state, the connecting mechanism **153** may be folded into the accommodation bin **152** to realize the storage of the connecting mechanism **153**.
(114) For example, when it is necessary to transport the target object between accommodation bin **152** and the target UAV, the connecting mechanism **153** may be unfolded to keep the connecting mechanism **153** in the unfolded state so as to achieve the transfer of the target object between the accommodation bin **152** and the target UAV. When the accommodation bin **152** needs to be raised or lowered, the connecting mechanism **153** may be folded into the accommodation bin **152** to avoid the connecting mechanism **153** from affecting the rising and falling of the accommodation bin **152**.
(115) In some exemplary embodiments, the takeoff and landing platform may also include: a base **16**. One end of the bracket **10** is fixed on the base **16**, and the other end of the bracket **10** extends in

a direction away from the base **16**. The bottom of the base **16** is also provided with an auxiliary moving device **17**. The auxiliary mobile device **17** may assist the movement of the takeoff and landing platform in order to improve the moving convenience of the takeoff and landing platform.
(116) In some exemplary embodiments, the auxiliary moving device **17** may include at least one of a roller or a caster. The present disclosure does not limit the auxiliary moving device **17**. In summary, the takeoff and landing platform described herein may at least have the following advantages:
(117) In the present disclosure, by providing a bracket on the takeoff and landing platform, in the case of landing of the multiple UAVs, the multiple UAVs may be vertically stacked on the bracket along the guide rail. In the case of take-off of the multiple UAVs, the multiple UAVs can take off from the bracket. That is, the takeoff and landing platform may be used to realize the takeoff and landing of the multiple UAVs. In this way, it is possible to avoid manual UAV take-off site layout, preparation and landing recovery operations, and reduce manpower and site investment when multiple UAVs perform collaborative operations. It may not only reduce the cost of multiple UAV collaborative operations, but also improve the efficiency of the multiple UAV collaborative operations.
(118) The present disclosure also provides a UAV. The UAV may include, but is not limited to, any one of passenger UAV, logistics UAV, aerial photography UAV, performance UAV, battle UAV and agricultural plant protection UAV. The present disclosure does not specifically limit the type of UAV.
(119) In some exemplary embodiments, the bottom of the UAV may be provided with one of a protruding structure and a recessing structure, and the top of the UAV forms the other one of the protruding structure and the recessing structure. The protruding structure may be at least partially embedded within the recessing structure to facilitate vertical stacking of multiple UAVs, so as to achieve high-density storage of multiple UAVs on the takeoff and landing platform.
(120) Specifically, a protruding structure can be provided at the bottom of the UAV, and a recessing structure may be provided at the top of the UAV. In the case of multiple UAVs stacked vertically, for two adjacent UAVs, the protruding structure at the bottom of the upper UAV may be at least partially embedded in the recessing structure at the top of the lower UAV. Alternatively, a recessing structure may be at the bottom of the UAV and a protruding structure may be at the top of the UAV. In the case of multiple UAVs stacked vertically, for two adjacent UAVs, the protruding structure at the top of the lower UAV may be at least partially embedded in the recessing structure at the bottom of the upper UAV.
(121) It should be noted that in the present disclosure, only the case where a protruding structure is provided at the bottom of the UAV and a recessing structure is provided at the top of the UAV is shown. The case where the recessing structure is at the bottom of the UAV and the protruding structure is at the top of the UAV may follow the same working principles.
(122) FIG. **15** is a schematic diagram of the structure of a UAV according to some exemplary embodiments of the present disclosure. FIG. **16** is a schematic diagram of the structure of the UAV shown in FIG. **15** viewed from another angle. FIG. **17** is a schematic diagram of the structure of an A-A section of the UAV shown in FIG. **15**.
(123) In some exemplary embodiments, the UAV may specifically include: a body **20** and a plurality of arms **21** connected to the body **20**. The recessing structure **22** may be arranged on the top of the body **20**, and the protruding structure **23** may be arranged on the bottom of the body **20**.
(124) Specifically, the body **20** of the UAV may serve as the structural main body of the UAV. One end of the arm **21** may be connected to the body **20**, and the other end thereof may be used to connect a propeller **25**. In practical applications, the arm **21** can be a folding machine arm or a non-folding machine arm. The present disclosure does not limit this.
(125) In the present disclosure, when the top of the body **20** is provided with a recessing structure **22** and the bottom of the body **20** is provided with a protruding structure **23**, for multiple UAVs

stacked vertically, for two UAVs adjacent to each other, the protruding structure **23** at the bottom of the upper UAV body **20** can be at least partially embedded in the recessing structure **22** at the top of the lower UAV body **20** structure to achieve assisted positioning and high-density storage of multiple UAVs.
(126) In some exemplary embodiments, the protruding structure **23** may include: a planar part **231** and a second guide part **232** arranged around the planar part **231**. The second guide part **232** may guide the protruding structure **23** into the recessing structure **22** of another UAV to facilitate vertical stacking of multiple UAVs.
(127) Specifically, the planar part **231** may be the bottom surface of the body **20**, and the second guide part **232** may be an inclined plane or a curved surface structure. Through the guiding function of the second guide part **232**, the planar part **231** can be guided into the recessing structure **22** on the top of another UAV. This allows the UAV to accurately land on the takeoff and landing platform and stack vertically to achieve high-density storage of the UAV.
(128) In some exemplary embodiments, a bearing platform **201** may also be provided on the top of the body **20**. The bearing platform **201** is located between a plurality of arms **21**. The bearing platform **201** may be used to cooperate with the planar part **231** of another UAV. The contact area between the bearing platform **201** and the planar part **231** is large, which may reduce the pressure between the upper and lower UAVs, thereby reducing wear and tear when UAVs are stacked.
(129) In the present disclosure, one end of the arm **21** is fixed on the bearing platform **201**, and the other end thereof extends in a direction away from the bearing platform **201** and inclined upward. A plurality of the arms **21** surrounds to form a recessing structure **22**.
(130) In a specific application, the UAV may generally include multiple arms **21**. Each arm **21** can be arranged around the bearing platform **201** on the top of the body **20**. In a direction from the bearing platform **201** to an outer edge of the body **20**, the arm **21** extends in an inclined upward direction. Therefore, a plurality of inclined and surrounding arms **21** can enclose to form a recessing structure **22** arranged around the bearing platform **201**. During the landing of the UAV, the protruding structure **23** at the bottom of the UAV body **20** may slide along the arm **21** to the bearing platform **201**; the UAV may accurately land onto the takeoff and landing platform and stacked vertically.
(131) It should be noted that in the drawings of the present disclosure, only the case where the UAV includes four arms **21** is shown. In practical applications, the number of arms **21** of the UAV can be set according to actual conditions. For example, the number of arms **21** may be 3, 5 or 8, etc. The present disclosure does not specifically limit the number of the arms **21** on the UAV.
(132) In some exemplary embodiments, a connecting frame **24** may also be provided between two adjacent arms **21**. The shape of the top surface of the connecting frame **24** is adapted to the shape of the top surface of the arm **21**. The connecting frame **24** and the arm **21** jointly form the recessing structure **22** to improve the integrity of the recessing structure **22** on the top of the UAV. In this way, during the landing process of the UAV, regardless the upper UAV lands on the arms **21** of the lower UAV or on the connecting frame **24** between the arms **21**, the above landing UAV can be guided onto the bearing platform **201** of the UAV below. This further improves the guiding range of the recessing structure **22** on the top of the UAV when the UAV lands, and improves the alignment efficiency when the UAV lands.
(133) It should be noted that, in order to reduce the overall weight of the UAV, the connection frame **24** may have a hollow design.
(134) In some exemplary embodiments, the connecting frame **24** is provided with a third opening **241** and a fourth opening **242**. The third opening **241** and the fourth opening **242** may be used to avoid guide rails on the takeoff and landing platform. In practical applications, when the UAV lands on the takeoff and landing platform, the third opening **241** may be sleeved over the guide rail of the first bracket, and the fourth opening **242** may be sleeved over the guide rail of the second bracket. In this way, on the one hand, during the landing process of the UAV, the guide rails on the

first bracket and the second bracket may guide the landing of the UAV, which is beneficial to the vertical stacking of the UAV. On the other hand, in the case where the multiple UAVs are vertically stacked on the takeoff and landing platform, by way of inserting the guide rail on the first bracket into the third opening **241** and inserting the guide rail on the second bracket into the fourth opening **242**, the guide rails may play a limiting role to prevent the UAV from swaying on the takeoff and landing platform in the circumferential direction. This further improves the stacking reliability of the multiple UAVs on the takeoff and landing platform.

(135) It should be noted that in the present disclosure, only the case where the UAVs are stacked on a takeoff and landing platform containing two brackets is shown. In specific applications, the number of openings on the connecting frame **24** may correspond to the number of brackets on the takeoff and landing platform. For example, where only one bracket is provided on the takeoff and landing platform, an opening can be adaptively provided on the connecting frame **24**; in the case where the takeoff and landing platform includes four brackets, four openings may be adaptively provided on the connecting frame **24**. The present disclosure does not limit the number of openings on the connecting frame **24**.

(136) In some exemplary embodiments, the UAV may also include: a propeller **25** and a propeller protective cover **26**. The propeller **25** is fixed on the other end of the arm **21**. The propeller protective cover **26** covers the propeller **25** and is used to protect the propeller **25**. A guide part is provided on a side of the propeller protective cover **26** close to the arm **21**. The guide part may guide the protruding structure **23** of another UAV into the recessing structure **22**.

(137) In specific applications, by providing the guide part on the side of the propeller protective cover **26** close to the arm **21**, the guide part may serve as a peripheral structure of the recessing structure **22**, which further increases the area of the recessing structure **22**. In this way, during the landing process of the UAV, whether the upper UAV lands on the arms **21** of the lower UAV or on the propeller protective cover **26** on the periphery of the arm **21**, the upper UAV may be guided to the bearing platform **201** of the lower UAV. This further improves the guiding range of the recessing structure **22** on the top of the UAV for another UAV to land, and improves the alignment efficiency of the UAV when it lands. In summary, the UAV described in the embodiments of the present disclosure may at least include the following advantages:

(138) In the present disclosure, by providing one of the protruding structure and the recessing structure at the bottom of the UAV, and the top of the UAV forms the other one of the protruding structure and the recessing structure, the protruding structure may be at least partially embedded within the recessing structure to facilitate vertical stacking of multiple UAVs. Moreover, in the case of vertical stacking of multiple UAVs, for two UAVs adjacent to each other, the protruding structure at the bottom of the upper UAV body may be at least partially embedded in the recessing structure at the top of the lower UAV body to achieve high-density storage of multiple UAVs.

(139) The present disclosure also provides a takeoff and landing (docking) system as shown in FIG. **2** to **4**. The takeoff and landing (docking) system may specifically include: the takeoff and landing platform as described in any of the above exemplary embodiments, and the UAV as described in any of the above exemplary embodiments, where there are multiple UAVs. The multiple UAVs can be stacked vertically on the takeoff and landing platform. In the case of landing of the multiple UAVs, the multiple UAVs can be vertically stacked on the takeoff and landing platform. In the case of takeoff of the multiple UAVs, the multiple UAVs can take off from the takeoff and landing platform.

(140) It should be noted that in the present disclosure, the structure of the takeoff and landing platform may be the same as the structure of the takeoff and landing platform in the above exemplary embodiments, and its working principle is also similar, which will not be described again herein. Similarly, the specific structure of the UAV may also be the same as the structure of the UAV in the above exemplary embodiments, and its working principle is also similar, which will not be described again herein.

(141) In the embodiments of the present disclosure, by providing a bracket on the takeoff and landing platform, when multiple UAVs land, the multiple UAVs may be vertically stacked on the bracket along the guide rail. In the case of takeoff of the multiple UAVs, the multiple UAVs may take off from the bracket. That is, the takeoff and landing platform may be used to realize the takeoff and landing of multiple UAVs. In this way, it is possible to avoid manual UAV takeoff site layout, preparation, and landing recovery operations, and reduce manpower and site investment when multiple UAVs perform collaborative operations. It can not only reduce the cost of multiple UAV collaborative operations, but also improve the efficiency of multiple UAV collaborative operations.
(142) The present disclosure also provides a storage device for storing the takeoff and landing platform and UAV.
(143) FIG. **18** is a schematic diagram of the structure of a storage device according to some exemplary embodiments of the present disclosure. FIG. **19** is a schematic diagram of the structure of the storage device shown in FIG. **18** viewed from another angle. FIG. **20** is a schematic diagram of the structure of the storage device shown in FIG. **18** viewed from another angle.
(144) Specifically, the storage device may include: a storage platform **30** and a takeoff and landing platform **31**. The number of takeoff and landing platforms **31** is multiple, and the multiple UAVs can be stacked vertically on each takeoff and landing platform **31**.
(145) In the present disclosure, the storage platform **30** may be used as the structural main body of the storage device to support multiple takeoff and landing platforms **31**. The takeoff and landing platform **31** can vertically stack multiple UAVs to fully utilize the vertical space to accommodate the multiple UAVs. In practical applications, by vertically stacking multiple UAVs on each takeoff and landing platform **31** and placing multiple takeoff and landing platforms **31** on the storage platform **30**, high-density storage of the UAVs can be achieved.
(146) In practical applications, multiple takeoff and landing platforms **31** can be placed on the storage platform **30** in the horizontal direction. Multiple UAVs on the takeoff and landing platform **31** can be stacked vertically on the bracket of the takeoff and landing platform **31**, so as to make full use of the horizontal space and vertical space of the storage device; high-density storage of the UAV is achieved, so as to facilitate the storage and transportation of the UAVs.
(147) FIG. **21** is a schematic diagram of the structure of a storage device according to some exemplary embodiments of the present disclosure. FIG. **22** is a schematic diagram of the structure of the storage device shown in FIG. **21** viewed from another angle.
(148) As shown in FIGS. **21** and **22**, a plurality of first telescopic frames **301** is provided on the storage platform **30**. The telescopic direction of the first telescopic frames **301** is the first direction. Each first telescopic frame **301** is connected with a multiple takeoff and landing platforms **31**. The multiple takeoff and landing platforms **31** are movably connected to the first telescopic frame **301** along the first direction to adjust the distance between adjacent takeoff and landing platforms **31**.
(149) In specific applications, the first telescopic frame **301** can be deployed along the first direction to facilitate connecting multiple lifting platforms to the first telescopic frame **301**. After multiple lifting platforms are connected to the first telescopic frame **301**, the first telescopic frame **301** can be retracted along the first direction, so that the distance between two adjacent takeoff and landing platforms **31** in the first direction is shortened, so as to reduce the space occupied by multiple takeoff and landing platforms **31** in the first direction.
(150) FIG. **23** is a schematic diagram of the structure of a first telescopic frame according to some exemplary embodiments of the present disclosure. As shown in FIG. **23**, the first telescopic frame **301** may include: a first frame body **3011** and a first telescopic mechanism **3012**. The first telescopic mechanism **3012** is telescopically connected to the first frame body **3011**. A plurality of takeoff and landing platforms **31** are connected to the first frame body **3011** to adjust the distance between adjacent takeoff and landing platforms **31** through the expansion and contraction of the first telescopic mechanism **3012**.

(151) Specifically, the first frame body **3011** can be used as a structural main body on the first telescopic bracket **301**, and the first frame body **3011** can be used to support the takeoff and landing platform **31**. The first telescopic mechanism **3012** can be movably connected to the first frame body **3011**. Multiple takeoff and landing platforms **31** can be connected to the first frame body **3011** via the first telescopic mechanism **3012**. In this way, through the expansion and contraction of the first telescopic mechanism **3012** along the first direction, the distance between two adjacent takeoff and landing platforms **31** in the first direction can be adjusted.
(152) For example, the first frame body **3011** may be a support rod extending along the first direction, and the first telescopic mechanism **3012** may be a spring, a folding hinge, etc. whose telescopic direction is along the first direction. The present disclosure does not limit the first frame body **3011** and the first telescopic mechanism **3012**.
(153) In some exemplary embodiments, the storage device may also include a second telescopic frame(s) **302**, the direction of the second telescopic frame **302** is a second direction, and the second direction is perpendicular to the first direction. A plurality of first telescopic frames **301** may be sequentially connected to the second telescopic frames **302** at intervals along the second direction to adjust the distance between adjacent first telescopic frames **301**.
(154) In the present disclosure, the first direction and the second direction may be perpendicular to each other. For example, a two-dimensional coordinate system may be established on the horizontal plane of the storage platform **30**, the first direction may be the horizontal axis direction in the two-dimensional coordinate system, and the second direction may be the vertical axis coordinate in the two-dimensional coordinate system. Alternatively, the first direction may be the vertical axis direction in the two-dimensional coordinate system, and the second direction may be the horizontal axis coordinate in the two-dimensional coordinate system. The present disclosure does not limit this.
(155) In specific applications, the second telescopic frame **302** may be deployed/expanded along the second direction to facilitate connecting multiple takeoff and landing platforms **31** to the second telescopic frame **301**. After multiple takeoff and landing platforms **31** are connected to the second telescopic frame **301**, the second telescopic frame **302** can be retracted along the second direction, so that the distance between two adjacent takeoff and landing platforms **31** in the second direction becomes shorter, so as to reduce the space occupied by the multiple takeoff and landing platforms **31** in the second direction.
(156) In some exemplary embodiments, the second telescopic frame **301** may include: a second frame body and a second telescopic mechanism. The second telescopic mechanism is telescopically connected to the second frame body. The plurality of first telescopic brackets may be connected to the second frame body to adjust the distance between adjacent second telescopic brackets through telescopic movement of the second telescopic mechanism.
(157) Specifically, the second frame body can be used as a structural main body of the telescopic frame, and the second frame body can be used to support the takeoff and landing platform **31**. The second telescopic mechanism can be movably connected to the second frame body, and the plurality of takeoff and landing platforms **31** can be connected to the second frame body via the second telescopic mechanism. In this way, through the expansion and contraction of the second telescopic mechanism along the second direction, the distance between the two adjacent takeoff and landing platforms **31** in the second direction may be adjusted.
(158) For example, the second frame body may be a support rod extending along the second direction. The second telescopic mechanism may be a spring, a folding hinge, etc. with a telescopic direction along the second direction. The present disclosure does not limit the specific second frame body and the second telescopic mechanism.
(159) Specifically, for the specific structure of the second telescopic frame **301**, reference may be made to the first telescopic frame **301** shown in FIG. **23**, and will not be described again herein.
(160) It should be noted that in specific applications, casters or rollers may be provided at the

bottom of the lifting platform to assist the movement of the lifting platform on the storage platform **30**, so the distance between the lifting platforms may be adjusted.
(161) In the present disclosure, the storage device may not only be used to store the UAVs, but may also be used for the takeoff and landing of multiple UAVs. Specifically, when the UAV needs to be stored or transported, the first telescopic frame **301** and the second telescopic frame **302** may be retracted, the distance between the takeoff and landing platforms **31** is shortened, and the space occupied by the takeoff and landing platform **31** on the storage platform **30** is reduced, so as to store as many UAVs as possible in the smallest possible space. When it is necessary to use the takeoff and landing platform **31** for UAV takeoff and landing, the first telescopic frame **301** and the second telescopic frame **302** may be deployed to increase the distance between the takeoff and landing platform **31**, so that a reasonable distance can be maintained between the takeoff and landing platform **31** to facilitate the takeoff or landing of the UAVs.
(162) In some exemplary embodiments, the UAVs may also be linked with the expansion and contraction of the takeoff and landing platform **31**. For example, when the distance between the takeoff and landing platform **31** becomes shorter, the UAVs on the takeoff and landing platform **31** may be automatically folded and stored to avoid interference between the UAVs on the takeoff and landing platform **31**. In another example, when the distance between the takeoff and landing platform **31** becomes larger, the UAVs on the takeoff and landing platform **31** may perform takeoff operations. Moreover, the UAVs on two adjacent takeoff and landing platforms **31** may take off at different times to provide the required headroom when the UAVs take off.
(163) In summary, the storage device according to the present disclosure may at least have the following advantages:
(164) In the present disclosure, the storage platform may be used as the structural body of the storage device to support multiple takeoff and landing platforms. The takeoff and landing platform can vertically stack multiple UAVs to fully utilize the vertical space to accommodate the multiple UAVs. In practical applications, by stacking multiple UAVs vertically on each takeoff and landing platform and placing multiple takeoff and landing platforms on the storage platform, high-density storage of the UAVs can be achieved.
(165) The present disclosure also provides a takeoff and landing control method, which is applied to the main control device.
(166) FIG. **24** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure. The takeoff and landing control method may specifically include the following steps:
(167) Step S**11**: Receive an operation instruction for a plurality of UAVs, where the operation instruction includes a takeoff instruction and a recall instruction.
(168) In the present disclosure, the main control device may be a device used to control the takeoff and landing platform and the UAV to perform related operations, The main control device includes but is not limited to any one of a computer, a console, or a remote control, and the present disclosure does not limit the specific main control device.
(169) Specifically, the communication connection between the main control device and the takeoff and landing platform may be realized through a wired connection or a wireless connection. The communication connection between the main control device and the UAV can be realized through wireless connection. Based on the communication connection between the main control device, the takeoff and landing platform and the UAV, the main control device may send control instructions to the takeoff and landing platform and the UAV to control the takeoff and landing platform or the UAV to perform relevant operations. In some exemplary embodiments, the main control device may be integrated into the takeoff and landing platform. In some exemplary embodiments, the main control device may be one of the multiple UAVs.
(170) In the present disclosure, the main control device may be provided with an instruction receiving device, and the instruction receiving device may be used for the operation instructions for

multiple UAVs. The instruction receiving device may include, but is not limited to, at least one of a touch screen, a button, or a voice recognition module. In specific applications, users may issue operation instructions for the UAV through touch operations on the touch screen, pressing or rotating operations on buttons, and by voice.
(171) Specifically, the operation instructions may include takeoff instructions and recall instructions. The takeoff instructions may be used to control the UAV to take off from the takeoff and landing platform; the recall command may be used to recall the UAV and vertically stack the plurality of UAVs on the bracket of the takeoff and landing platform.
(172) Step S**12**: In response to the takeoff instruction, control the plurality of UAVs stacked vertically on the takeoff and landing platform to take off from the takeoff and landing platform.
(173) In the present disclosure, after receiving a takeoff instruction for the UAVs, the main control device may control multiple UAVs vertically stacked on the takeoff and landing platform to take off from the takeoff and landing platform in response to the takeoff instruction. In specific applications, when there are multiple UAVs stacked vertically on the takeoff and landing platform, the main control device may control the multiple UAVs to take off in sequence from top to bottom.
(174) Specifically, the takeoff and landing platform may be provided with a first charging module. The first charging module is electrically connected to the second charging module on the UAV to charge the UAV. In practical applications, before controlling the UAVs to take off, the UAVs should be controlled to release/disconnect the electrical connection with the first charging module, in order to facilitate the UAVs to perform takeoff operations and take off from the takeoff and landing platform.
(175) In some exemplary embodiments, the method of controlling multiple UAVs stacked vertically on a takeoff and landing platform to take off from the takeoff and landing platform may specifically include the following sub-steps:
(176) First, control the plurality of UAVs stacked vertically on the takeoff and landing platform to rotate their propellers to a hovering state.
(177) In the present disclosure, when it is necessary to control the plurality of UAVs on the takeoff and landing platform to take off, the propellers of the plurality of UAVs stacked vertically on the takeoff and landing platform may be controlled to start rotating to provide a lifting force to the UAVs. Specifically, before taking off from the takeoff and landing platform, all UAVs on the takeoff and landing platform can be controlled to rotate their propellers to a hovering state. In practical applications, when the UAVs are in the hovering state, the UAVs are in a state that can take off.
(178) Specifically, the hovering state may be as follows: the lifting force generated by the rotation of the propellers on the UAV is not enough to enable the UAV to complete a full take-off action; however, the UAV can maintain a stable attitude (the attitude may include a horizontal attitude and an attitude in a rotational direction) and move along the guidance feature with a point of support below.
(179) In the present disclosure, by controlling the propellers of the plurality of UAVs stacked vertically on the takeoff and landing platform to the hovering state, the plurality of UAVs on the takeoff and landing platform may be controlled in a takeoff-ready state.
(180) Next, the takeoff and landing platform is controlled to push the multiple UAVs out of the takeoff and landing platform in sequence, and control the pushed-out UAVs to take off.
(181) In the present disclosure, after controlling the plurality of UAVs on the takeoff and landing platform to rotate the propellers thereof to reach the hovering state, the main control device may control the takeoff and landing platform based on the communication connection between the main control device and the takeoff and landing platform to push the multiple UAVs out of the takeoff and landing platform in sequence, and based on the communication connection between the main control device and the UAV, control the pushed-out UAVs to take off.
(182) Specifically, after controlling the plurality of UAVs on the takeoff and landing platform to

rotate the propellers thereof to reach the hovering state, the main control device may control the lifting module on the takeoff and landing platform to lift the plurality of UAVs by the height of one UAV, so the uppermost UAV is pushed out of the takeoff and landing platform, and then the pushed-out UAV is controlled to take off. Next, the above steps are performed in a loop in an order of from top to bottom; the UAVs on the takeoff and landing platform are sequentially pushed out from the takeoff and landing platform and to take off. This enables the plurality of UAVs on the takeoff and landing platform to be pushed out in sequence to take off one after another, thereby achieving continuous takeoff to reach their destination.
(183) In the present disclosure, by the method including controlling the plurality of UAVs stacked vertically on the takeoff and landing platform to rotate the propellers thereof to reach a hovering state, then controlling the takeoff and landing platform to push the plurality of UAVs out of the takeoff and landing platform in sequence, and controlling the pushed out UAV to take off, since the UAV is already in a hovering state on the takeoff and landing platform and is ready to take off, after being pushed out from the takeoff and landing platform, the UAV can take off quickly, so that the plurality of UAVs may take off continuously, and the takeoff efficiency is high.
(184) In some exemplary embodiments, the controlling of the plurality of UAVs stacked vertically on a takeoff and landing platform to take off from the takeoff and landing platform may specifically include the following sub-steps:
(185) First, control the takeoff and landing platform to push out the UAVs stacked vertically in a propeller stopping state.
(186) In some exemplary embodiments, when it is necessary to control the takeoff of the plurality of UAVs from the takeoff and landing platform, the takeoff and landing platform may be controlled first to push the topmost UAV among the UAVs that are vertically stacked in the propeller stopping state out from the takeoff and landing platform.
(187) Specifically, the main control device may control the lifting module of the takeoff and landing platform to lift the UAV by the height of the body, so as to push all UAVs on top of the takeoff and landing platform out of the takeoff and landing platform.
(188) Next, control the pushed out UAV to take off.
(189) In the present disclosure, after pushing the uppermost UAV on the takeoff and landing platform out from the takeoff and landing platform, the main control device may control the propellers of the pushed out UAV to rotate to take off. Then, the above steps are performed in a loop until all UAVs on the takeoff and landing platform are taken off in an order from top to bottom.
(190) Specifically, since the UAV controls the UAV's propellers to rotate after the UAV is pushed out from the takeoff and landing platform, this can prevent the UAV from rotating its propellers on the takeoff and landing platform, so as to avoid interference between the UAV propellers and the takeoff and landing platform to the greatest extent, and improve the takeoff safety of the UAV.
(191) In some exemplary embodiments, the controlling of the plurality of UAVs stacked vertically on the takeoff and landing platform to take off from the takeoff and landing platform may specifically include the following sub-steps:
(192) First, control the takeoff and landing platform to push out the plurality of vertically stacked UAVs.
(193) In the present disclosure, when it is necessary to control the plurality of UAVs on the takeoff and landing platform to take off, the main control device may first control the takeoff and landing platform to push all UAVs vertically stacked on the takeoff and landing platform out of the takeoff and landing platform.
(194) Specifically, the main control device may control the lifting module of the takeoff and landing platform to lift the plurality of UAVs to reach a higher height, until all UAVs on the takeoff and landing platform are pushed out of the takeoff and landing platform. Then, the plurality of UAVs may be controlled to rotate the propellers thereof to reach a hovering state.

(195) In the present disclosure, after all UAVs on the takeoff and landing platform are pushed out of the takeoff and landing platform, the main control device shown may control the propellers of the plurality of UAVs to rotate, so that the plurality of UAVs rotate the propellers thereof to reach a hovering state. That is, the plurality of UAVs pushed out the takeoff and landing platform are placed in a takeoff-ready state.
(196) Finally, the plurality of UAVs in the hovering state is controlled to take off simultaneously.
(197) In the present disclosure, when the plurality of UAVs is pushed out of the takeoff and landing platform and rotate their propellers to reach a hovering state, the main control device may control the plurality of UAVs in the hovering state to take off at the same time, and reach their target locations, thus the takeoff efficiency is extremely high.
(198) Specifically, the takeoff and landing platform is controlled to push out the plurality of vertically stacked UAVs, the plurality of UAVs are controlled to rotate their propellers to reach a hovering state, and finally the plurality of UAVs in the hovering state are controlled to take off at the same time. This may prevent the UAVs from rotating their propellers on the takeoff and landing platform, so as to avoid interference between the UAV propellers and the takeoff and landing platform to the greatest extent, and improve the takeoff safety of the UAVs. Moreover, it can also achieve higher take-off efficiency.
(199) It should be noted that in specific applications, a person skilled in the art may use the method described in any of the above exemplary embodiments to control multiple UAVs stacked vertically on the takeoff and landing platform to take off. The present disclosure does not specifically limit the takeoff method of the UAVs from the takeoff and landing platform.
(200) Step S**13**: In response to the recall instruction received, control the plurality of UAVs to land and be stacked vertically on the takeoff and landing platform along the guide rail.
(201) In the present disclosure, after the main control device receives the recall instruction for the UAV, in response to the recall instruction, the plurality of UAVs may be controlled to land and be stacked vertically on the takeoff and landing platform along the guide rail. In specific applications, when recalling the multiple UAVs, the first recalled UAV may be stacked at the bottom of the takeoff and landing platform.
(202) In some exemplary embodiments, the controlling of the plurality of UAVs to land and be stacked vertically on the takeoff and landing platform along the guide rail may specifically include the following sub-steps:
(203) First, recall the UAVs to the top of the takeoff and landing platform and switch to the hovering state.
(204) In the present disclosure, when the UAVs need to be recalled, the main control device may control a first UAV to land within the range of the guide carrier plate of the takeoff and landing platform through precise landing, and reduce the propeller speed of the UAV, causing the UAV to switch from a flying state to a hovering state.
(205) In practical applications, a first identity mark may be provided on the guide carrier plate of the takeoff and landing platform. During the landing process, the UAV may accurately land within the range of the guide carrier board by identifying the first identity mark on the guide carrier board. The first identity mark may include, but is not limited to, image marks such as QR codes, or contour features such as protrusions or recesses. The present disclosure does not specifically limit the first identity identifier.
(206) Next, after each UAV is determined to be in a hovering state, the UAVs are controlled to land and stack vertically on the takeoff and landing platform along the guide rail.
(207) In the present disclosure, after determining that one of the UAVs has landed within the range of the guide carrier plate of the takeoff and landing platform and is in a hovering state, the UAV may be controlled to further descend and stack on the takeoff and landing platform along the guide rail.
(208) In practical applications, since the guide carrier plate includes a lower recess part and a first

guide part, the lower recess part is arranged in the central area of the guide carrier plate, and the first guide part is arranged in the edge area of the guide carrier plate and surrounds the lower recess part. When the UAV lands on the guide carrier plate is in a hovering state, the bottom of the UAV may slide along the first guide part until the UAV lands on the lower recess part in the central area of the guide carrier plate to complete the landing process of the UAV.

(209) In the present disclosure, after the first UAV lands on the guide carrier plate of the takeoff and landing platform, the main control device may control the guide carrier plate to lower by the height of the UAV body to recover the UAV on the guide carrier plate into the takeoff and landing platform. Next, the above steps are repeated to recall the second UAV to the range of the takeoff and landing platform and control the UAV to switch to the hovering state.

(210) Specifically, during the landing process of the second UAV, the position of the first UAV that has landed on the takeoff and landing platform can be referenced to perform precise landing. In practical applications, a second identity mark can be provided on each UAV. By identifying the second identity mark on the UAV, another UAV may be controlled to accurately land within the range of the UAV. The second identity mark may include, but is not limited to, image marks such as QR codes, or contour features such as protrusions or recesses. The present disclosure does not specifically limit the second identity mark.

(211) In practical applications, since the protruding structure is set at the bottom of the UAV and the recessing structure is set at the top of the UAV, in the event that the bottom of the second UAV lands within range of the first UAV and is in a hovering state, the protruding structure at the bottom of the upper UAV may slide along the recessing structure at the top of the lower UAV until the upper UAV lands on the bearing platform at the top of the lower UAV to complete the landing process of the second UAV.

(212) In the present disclosure, after the second UAV lands on the first UAV on the takeoff and landing platform, the main control device may control the guide carrier plate to lower by the height of the UAV body to recover the second UAV to the takeoff and landing platform. Next, the landing steps of the second UAV are repeated until all UAVs are recalled and stacked vertically on the takeoff and landing platform along the guide rail.

(213) In some exemplary embodiments, the controlling of the plurality of UAVs to land and stack vertically on the takeoff and landing platform along the guide rail may specifically include the following sub-steps:

(214) First, recall the plurality of UAVs simultaneously to the top of the takeoff and landing platform and switch to a hovering state and arranged vertically in the vertical direction.

(215) In the present disclosure, when the UAV needs to be recalled, the main control device can simultaneously recall the plurality of UAVs to the top of the takeoff and landing platform, arrange the plurality of UAVs in a vertical direction in an order of from top to bottom, and arrange the plurality of UAVs in a vertical direction; the UAV switches to the hovering state.

(216) Specifically, in practical applications, the UAV arranged at the bottom may land within the range of the guide carrier plate of the takeoff and landing platform by precise landing. For specific implementation methods, reference may be made to the descriptions in the foregoing exemplary embodiments and will not be described again herein. Counting from bottom to top, the second UAV may accurately land on the top of the first UAV based on the position of the first UAV. For specific implementation methods, reference may also be made to the descriptions in the foregoing exemplary embodiments and will not be described again herein, and so on, until the UAVs are arranged vertically on the takeoff and landing platform, and all UAVs are switched to the hovering state.

(217) In the present disclosure, the UAV arranged at the bottom can land on the guide carrier plate of the takeoff and landing platform. The guide carrier plate includes a lower recess part and a first guide part. The lower recess part is arranged in the central area of the guide carrier plate, and the first guide part is disposed at an edge area of the guide carrier plate and surrounds the lower recess

part. When the UAV landed on the guide carrier plate is in a hovering state, the bottom of the UAV may slide along the first guide part until the UAV lands on the lower recess part in the central area of the guide carrier plate to complete the landing process of the UAV.
(218) For two UAVs adjacent to each other, the upper UAV may land on the top of the lower UAV. A protruding structure is provided at the bottom of the UAV, and a recessing structure is set at the top of the UAV. Therefore, in the case where the bottom of the upper UAV comes within range of the lower UAV and is in a hovering state, the protruding structure at the bottom of the upper UAV can slide along the recessing structure at the top of the lower UAV until the upper UAV lands on the bearing platform at the top of the lower UAV, completing the landing of the upper UAV. By executing the above method in a loop, the plurality of UAVs may be vertically stacked above the takeoff and landing platform.
(219) For example, in the case where 16 UAVs need to be stacked vertically on the takeoff and landing platform, these 16 UAVs can be recalled to the top of the takeoff and landing platform at the same time. The 16 UAVs may be arranged vertically in a vertical direction and can be switched to a hovering state.
(220) Next, the multiple UAVs in the hovering state are controlled to land sequentially and stacked vertically on the takeoff and landing platform along the guide rail.
(221) In the present disclosure, when the main control device controls the plurality of UAVs to be arranged vertically in the vertical direction and switched to the hovering state, the main control device may control the guide carrier plate of the takeoff and landing platform to descend to simultaneously recover the multiple UAVs to the takeoff and landing platform.
(222) It should be noted that in specific applications, a person skilled in the art can recall multiple UAVs according to the method described in any of the above exemplary embodiments, and control the multiple UAVs to be vertically stacked on the takeoff and landing platform. The present disclosure does not specifically limit the manner in which the UAV lands on the takeoff and landing platform.
(223) FIG. **25** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure. As shown in FIG. **25**, after step S**13**, the takeoff and landing control method may also include the following steps:
(224) Step S**14**: In response to a received storage instruction, control the takeoff and landing platform to store the plurality of vertically stacked UAVs.
(225) In the present disclosure, the operation instruction may also include a storage instruction. When stacking the plurality of UAVs vertically on the takeoff and landing platform, the main control device may receive the storage instruction; in response to the storage instruction, the takeoff and landing platform may be controlled to store the plurality of vertically stacked UAVs so as to store the plurality of UAVs onto the takeoff and landing platform. The space occupied by the plurality of UAVs on the takeoff and landing platform may be minimized.
(226) Specifically, the main control device may control the guide carrier plate on the takeoff and landing platform to descend to the lowest point of the takeoff and landing platform, so as to achieve the storage of the plurality of UAVs and facilitate the transportation of the takeoff and landing platform and the plurality of UAVs on it.
(227) Step S**15**: In response to a received charging instruction, control the takeoff and landing platform to charge the plurality of UAVs.
(228) In the present disclosure, the storage instruction may also include a charging instruction. When stacking the plurality of UAVs vertically on the takeoff and landing platform, the main control device receives the charging instruction. In response to the charging instruction, the takeoff and landing platform may be controlled to charge the plurality of vertically stacked UAVs to improve the endurance of the plurality of UAVs on the takeoff and landing platform.
(229) Specifically, the takeoff and landing platform may be provided with a first charging module(s), and one of the first charging module(s) may be electrically connected to a second

charging module of the UAV to charge the UAV.
(230) In the present disclosure, the body height of the UAV is the first height; the vertical interval between two adjacent first charging modules is a first interval, and the first interval is greater than the first height. Therefore, in the event that the UAV on the takeoff and landing platform needs to be charged, the interval between two adjacent UAVs may be adjusted so that the interval between two adjacent UAVs is equal to the first interval, so that the first charging module on the takeoff and landing platform and the second charging module on the UAV are aligned one by one to facilitate electrical connection between the two.
(231) In summary, the takeoff and landing control method described in the exemplary embodiment of the present disclosure may specifically include the following advantages:
(232) In the present disclosure, the main control device may receive an operation instruction for the plurality of UAVs, where the operation instruction may include a takeoff instruction and a recall instruction. In response to the takeoff instruction, the plurality of UAVs stacked vertically on the takeoff and landing platform is controlled to take off from the takeoff and landing platform. In response to the received recall instruction, the plurality of UAVs are controlled to land and vertically stack on the takeoff and landing platform along the guide rail to achieve takeoff and landing of the plurality of UAVs on the takeoff and landing platform. In this way, it is possible to avoid manual UAV take-off site layout, preparation, and landing recovery operations, and reduce manpower and site investment when multiple UAVs perform collaborative operations. It can not only reduce the cost of multiple UAV collaborative operations, but also improve the efficiency of multiple UAV collaborative operations.
(233) The present disclosure also provides a takeoff and landing control method, which can be used for the takeoff and landing platform described in the above exemplary embodiments.
(234) FIG. **26** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure. As shown in FIG. **26**, the takeoff and landing control method may specifically include the following steps:
(235) Step S**21**: In response to the received takeoff instruction, push out the plurality of UAVs vertically stacked on the takeoff and landing platform from the takeoff and landing platform.
(236) In the present disclosure, after receiving the operation instruction for the UAV, the main control device may send the operation instruction to the takeoff and landing platform. Specifically, the operation instruction may include a takeoff instruction and a recall instruction. The takeoff instruction may be used to control the UAV to take off from the takeoff and landing platform; the recall instruction may be used to recall the UAV and vertically stack a plurality of UAVs on the bracket of the takeoff and landing platform.
(237) In practical applications, after receiving the takeoff instruction sent by the main control device, the takeoff and landing platform may control the guide carrier plate to move upward to push out multiple UAVs vertically stacked on the takeoff and landing platform from the takeoff and landing platform, so as to facilitate the takeoff of the multiple UAVs from the takeoff and landing platform.
(238) It should be noted that the speed at which the takeoff and landing platform push out the UAV needs to be determined based on the takeoff mode of the UAV. For details, reference may be made to the foregoing exemplary embodiments, and the present disclosure does not describe it in detail.
(239) Step S**22**: In response to the received recall instruction, vertically stack the landed UAVs on the takeoff and landing platform along the guide rail.
(240) In the present disclosure, after receiving the recall instruction sent by the main control device, the takeoff and landing platform, after the plurality of UAVs land on the guide carrier plate, control the guide carrier plate to move downward to stack the plurality of UAVs vertically on the takeoff and landing platform along the guide rail of the takeoff and landing platform, so as to land the plurality of UAVs onto the takeoff and landing platform.
(241) It should be noted that the speed at which the takeoff and landing platform takes back the

UAV needs to be determined according to the recall mode of the UAV. For details, reference may be made to the aforementioned exemplary embodiments. The present disclosure does not elaborate on this.

(242) FIG. **27** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure. As shown in FIG. **27**, after step **22**, the takeoff and landing control method may also include the following steps:

(243) Step S**23**: In response to a received storage instruction, store the plurality of UAVs vertically stacked on the takeoff and landing platform.

(244) In the present disclosure, the operation instruction may also include a storage instruction. In the case of stacking the plurality of UAVs vertically on the takeoff and landing platform, when the takeoff and landing platform receives a storage instruction sent by the main control device, in response to the storage instruction, the plurality of vertically stacked UAVs may be controlled to be stored, so as to store the plurality of UAVs onto the takeoff and landing platform. Thus, the space occupied by the plurality of UAVs on the takeoff and landing platform is minimized.

(245) Specifically, when the takeoff and landing platform receives the takeoff instruction sent by the main control device, the takeoff and landing platform may be controlled to lower the vertically stacked plurality of UAVs to a storage position to achieve storage of the plurality of UAVs, and facilitate transportation of the takeoff and landing platform and the plurality of UAVs on it.

(246) For example, the storage position may be a position where the guide carrier plate descends to the lowest point of the takeoff and landing platform.

(247) Step S**24**: In response to a received charging instruction, charge the plurality of UAVs vertically stacked on the takeoff and landing platform.

(248) In the present disclosure, the storage instruction may also include a charging instruction. When stacking the plurality of UAVs vertically on the takeoff and landing platform, the main control device may receive a charging instruction, and the takeoff and landing platform may be controlled to charge the plurality of vertically stacked UAVs in response to the charging instruction, so as to improve the endurance of the plurality of UAVs on the takeoff and landing platform.

(249) Specifically, the takeoff and landing platform may be provided with a first charging module(s), and one of the first charging modules may be electrically connected to a second charging module of the UAV to charge the UAV.

(250) In the present disclosure, the height of the UAV's body is the first height, and the vertical interval between two adjacent first charging modules is the first interval. The first interval is greater than the first height. Therefore, in the event that the UAV on the takeoff and landing platform needs to be charged, the interval between two adjacent UAVs may be adjusted so that the interval between two adjacent UAVs is equal to the first interval, so that the first charging module on the takeoff and landing platform and the second charging module on the UAV may be aligned one by one to facilitate electrical connection between the two.

(251) In summary, the takeoff and landing control method described in the present disclosure may at least have the following advantages:

(252) In the present disclosure, the takeoff and landing platform may respond to a received takeoff instruction by pushing out the plurality of UAVs vertically stacked on the takeoff and landing platform from the takeoff and landing platform, and in response to a received recall instruction, vertically stack the landed UAVs on the takeoff and landing platform along the guide rail, so as to achieve takeoff and landing of the plurality of UAVs on the takeoff and landing platform. In this way, it is possible to avoid manual UAV take-off site layout, preparation, and landing recovery operations, and reduce manpower and site investment when multiple UAVs perform collaborative operations. It may not only reduce the cost of multiple UAV collaborative operations, but also improve the efficiency of multiple UAV collaborative operations.

(253) The present disclosure further provides a takeoff and landing control method, which may be applied to the UAVs in the above exemplary embodiments.

(254) FIG. **28** is a schematic diagram of a step flow chart of a takeoff and landing control method according to some exemplary embodiments of the present disclosure. As shown in FIG. **28**, the takeoff and landing control method may specifically include the following steps:
(255) Step S**31**: In response to the received takeoff instruction, the plurality of UAVs takes off from the takeoff and landing platform.
(256) In the present disclosure, after receiving the operation instruction for the UAV, the main control device may send the operation instruction to the UAV. Specifically, the operation instruction may include a takeoff instruction and a recall instruction. The takeoff instruction be used to control the UAV to take off from the takeoff and landing platform, and the recall instruction may be used to recall the UAV and vertically stack the plurality of UAVs on the bracket of the takeoff and landing platform.
(257) In practical applications, the UAV takes off from the takeoff and landing platform after receiving a takeoff instruction sent by the main control device.
(258) In some exemplary embodiments, the method for the plurality of UAVs to take off from the takeoff and landing platform may include the following sub-steps:
(259) First, the plurality of UAVs stacked vertically on the takeoff and landing platform rotate their propellers to reach a hovering state.
(260) In the present disclosure, under the control of the main control device, the propellers of the plurality of UAVs stacked vertically on the takeoff and landing platform can may to rotate to provide a lifting force for the UAV. Specifically, before flying out of the takeoff and landing platform, all UAVs on the takeoff and landing platform may rotate to reach a hovering state. In practical applications, when the UAV is in the hovering state, the UAV is in a state ready to take off.
(261) Next, as pushed by the takeoff and landing platform, the plurality of UAVs is pushed out from the takeoff and landing platform in sequence, and the pushed-out UAVs may take off.
(262) In the present disclosure, after the plurality of UAVs on the takeoff and landing platform rotate the propellers thereof to reach the hovering state, the takeoff and landing platform may push the plurality of UAVs out of the takeoff and landing platform in sequence. Under the influence of the takeoff and landing platform, the plurality of UAVs is pushed out from the takeoff and landing platform in sequence, and the pushed-out UAVs may take off.
(263) Specifically, after the plurality of UAVs on the takeoff and landing platform rotate the propellers thereof to reach the hovering state, the lifting module on the takeoff and landing platform may lift the plurality of UAVs by the height of one UAV to push the uppermost UAV out of the takeoff and landing platform. The takeoff operation may be performed after the UAV is pushed out of the takeoff and landing platform. Next, the above steps is performed repeatedly, in an order of from top to bottom, the UAVs on the takeoff and landing platform are pushed out of the takeoff and landing platform and then perform the takeoff operation. This enables the plurality of UAVs on the takeoff and landing platform to be pushed out and take off one after another, achieving continuous takeoff and reaching the destination.
(264) In the present disclosure, since the UAV is already in a hovering state on the takeoff and landing platform and can take off, after being pushed out from the takeoff and landing platform, the UAV can take off quickly. In this way, the continuous takeoff of the plurality of UAVs can be achieved, and the takeoff efficiency is high.
(265) In some exemplary embodiments, the method for the plurality of UAVs to take off from the takeoff and landing platform may include the following sub-steps:
(266) First, the UAVs that are vertically stacked and in a propeller stopping state are pushed out by the takeoff and landing platform.
(267) In the present disclosure, the takeoff and landing platform can be used to push the uppermost UAV among the UAVs that are vertically stacked and in the propeller stopping state out of the takeoff and landing platform.

(268) Specifically, the lifting module on the takeoff and landing platform may lift the UAV by the height of one UAV body to push all the uppermost UAVs on the takeoff and landing platform out of the takeoff and landing platform.
(269) The pushed-out UAVs then take off.
(270) In the present disclosure, after pushing the uppermost UAVs on the takeoff and landing platform out of the takeoff and landing platform, the propellers of these UAVs rotate and the UAVs take off. Then, the above steps are performed in a loop until all UAVs on the takeoff and landing platform take off in sequence from top to bottom.
(271) Specifically, since the UAVs are pushed out of the takeoff and landing platform and then rotate their propellers, the operation of the UAV rotating the propellers on the takeoff and landing platform can be avoided, so as to avoid interference between the propellers of the UAV and the takeoff and landing platform to the greatest extent, the takeoff safety of the UAV is therefore improved.
(272) In some exemplary embodiments of the present disclosure, the method for the plurality of UAVs to take off from the takeoff and landing platform may include the following sub-steps:
(273) First, under the push-out effect of the takeoff and landing platform, the plurality of vertically stacked UAVs are pushed out of the takeoff and landing platform.
(274) In the present disclosure, the takeoff and landing platform pushes all UAVs vertically stacked on the takeoff and landing platform out of the takeoff and landing platform.
(275) Specifically, the lifting module on the takeoff and landing platform may lift the plurality of UAVs higher until all the UAVs on the takeoff and landing platform are pushed out of the takeoff and landing platform.
(276) Next, the plurality of UAVs rotates the propellers thereof to reach a hovering state.
(277) In the present disclosure, after all UAVs on the takeoff and landing platform are pushed out of the takeoff and landing platform, the propellers of the plurality of UAVs rotate, so that the plurality of UAVs reach a hovering state. That is, the plurality of UAVs pushed out of the takeoff and landing platform is in a state ready to take off.
(278) Finally, the plurality of UAVs in the hovering state takes off simultaneously.
(279) In the present disclosure, when the plurality of UAVs are pushed out of the takeoff and landing platform and rotate the propellers thereof to reach the hovering state, the plurality of UAVs may take off at the same time to reach the target location(s) thereof, and the take-off effect is therefore extremely high.
(280) Specifically, by means of controlling the takeoff and landing platform to push out the plurality of vertically stacked UAVs, controlling the plurality of UAVs to rotate the propellers to reach a hovering state, and finally controlling the plurality of UAVs in the hovering state to take off simultaneously, the present disclosure may prevent the UAV from rotating the propellers on the takeoff and landing platform, so as to avoid interference between the UAV propellers and the takeoff and landing platform to the greatest extent. This may improve the take-off safety of the UAVs and achieve higher take-off efficiency.
(281) It should be noted that the take-off speed of the UAV needs to be determined according to the take-off mode of the UAV. For details, reference may be made to the foregoing exemplary embodiments, which will not be described again herein.
(282) Step S**32**: In response to a received recall instruction, the plurality of UAVs land and are stacked vertically on the takeoff and landing platform along the guide rail.
(283) In the present disclosure, after the UAVs receive the recall instruction sent by the main control device, the UAVs land on the guide carrier plate and are then vertically stacked on the takeoff and landing platform along the guide rail of the takeoff and landing platform, so as to land on the takeoff and landing platform.
(284) In some exemplary embodiments of the present disclosure, the method for the plurality of UAVs to land on the takeoff and landing platform may include the following sub-steps:

(285) First, the UAVs are recalled to the top of the takeoff and landing platform and switched to the hovering state.
(286) In the present disclosure, when the UAVs need to be recalled, the main control device may control the first UAV to land within the range of the guide carrier plate of the takeoff and landing platform through precise landing, and then reduce the rotation speed of the propellers of the UAV, causing the UAV to switch from a flight state to a hovering state.
(287) In practical applications, a first identity mark may be provided on the guide carrier plate of the takeoff and landing platform. During the landing process, the UAV may precisely land within the range of the guide carrier board by identifying the first identity mark on the guide carrier board. The first identity mark may include, but is not limited to, image marks such as QR codes, or contour features such as protrusions or recesses. The present disclosure does not specifically limit the first identity mark.
(288) Next, after each UAV is confirmed to be in the hovering state, the UAVs in the hovering state land and are stacked vertically on the takeoff and landing platform along the guide rail.
(289) In the present disclosure, after determining that one of the UAVs has landed within the range of the guide carrier plate of the takeoff and landing platform and is in a hovering state, the UAV may be controlled to further land and be stacked on the takeoff and landing platform along the guide rail.
(290) Specifically, after the first UAV lands on the guide carrier plate of the takeoff and landing platform, the main control device may control the guide carrier plate to lower by the height of the UAV body to recover the UAV on the guide carrier plate to the takeoff and landing platform. Next, the above steps are repeated to recall the second UAV to the range of the takeoff and landing platform and control this UAV to switch to the hovering state.
(291) Specifically, during the landing process of the second UAV, the position of the first UAV that has landed on the takeoff and landing platform can be referenced to perform precise landing. In practical applications, a second identity may be provided on each UAV. By identifying the second identity mark on the UAV, another UAV may be controlled to precisely land within the range of this UAV. The second identity mark may include, but is not limited to, image marks such as QR codes, or contour features such as protrusions or recesses. The present disclosure does not specifically limit the second identity mark.
(292) In practical applications, since the protruding structure is arranged at the bottom of the UAV and the recessing structure is provided at the top of the UAV, in the event that the bottom of the second UAV lands in the range of the first UAV and is in a hovering state, the protruding structure at the bottom of the upper UAV may slide along the recessing structure at the top of the lower UAV until the upper UAV lands on the bearing platform at the top of the lower UAV to complete the landing of the second UAV.
(293) In the present disclosure, after the second UAV lands on the first UAV on the takeoff and landing platform, the main control device may control the guide carrier plate to lower by the height of one UAV body to recover the second UAV to the takeoff and landing platform. Next, the landing steps of the second UAV are repeated until all UAVs are recalled and stacked vertically on the takeoff and landing platform along the guide rail.
(294) In some exemplary embodiments of the present disclosure, the method for the plurality of UAVs to land on the takeoff and landing platform may include the following sub-steps:
(295) First, the plurality of UAVs are simultaneously recalled to the top of the takeoff and landing platform, switched to a hovering state and arranged vertically in the vertical direction.
(296) In the present disclosure, when the UAVs need to be recalled, the plurality of UAVs may be recalled to the top of the takeoff and landing platform at the same time and arranged vertically in order from top to bottom; thereafter, the plurality of UAVs may switch to the hovering state.
(297) Specifically, in practical applications, the UAV arranged at the bottom may land within the range of the guide carrier plate of the takeoff and landing platform through precise landing. For

specific implementation methods, reference may be made to the descriptions in the foregoing exemplary embodiments and will not be described again herein. Counting from bottom to top, the second UAV may precisely land on the top of the first UAV according to the position of the first UAV. For specific implementation methods, reference can also be made to the descriptions in the foregoing exemplary embodiments and will not be described again herein, and so on, until the UAVs are arranged vertically on the top of the takeoff and landing platform, and all UAVs are switched to the hovering state.
(298) In the present disclosure, the UAV arranged at the bottom may land on the guide carrier plate of the takeoff and landing platform. The guide carrier plate includes a lower recess part and a first guide part, the lower recess part is arranged in the central area of the guide carrier plate, and the first guide part is arranged in the edge area of the guide carrier plate and surrounds the lower recess part. Thus, when the UAV landing on the guide carrier plate is in a hovering state, the bottom of the UAV may slide along the first guide part until the UAV lands on the lower recess part in the central area of the guide carrier plate to complete the landing of the UAV.
(299) For two UAVs adjacent to each other, the upper UAV may land on the top of the lower UAV. Since the protruding structure is provided at the bottom of the UAV and the recessing structure is set at the top of the UAV, in the case where the bottom of the upper UAV comes within the range of the lower UAV and is in a hovering state, the protruding structure at the bottom of the upper UAV may slide along the recessing structure at the top of the lower UAV until the upper UAV lands on the bearing platform at the top of the lower UAV, thereby completing the landing of the upper UAV. By executing the above method in a loop, the plurality of UAVs may be vertically stacked on the takeoff and landing platform.
(300) For example, in the case where 16 UAVs need to be stacked vertically on the takeoff and landing platform, these 16 UAVs may be recalled simultaneously to the top of the takeoff and landing platform, and the 16 UAVs may be arranged vertically in a vertical direction and switched to a hovering state.
(301) Next, these UAVs in the hovering state land sequentially and are vertically stacked on the takeoff and landing platform along the guide rail.
(302) In the present disclosure, when the plurality of UAVs are vertically arranged in the vertical direction and switched to the hovering state, the guide carrier plate of the takeoff and landing platform may be lowered to simultaneously recover the plurality of UAVs to the takeoff and landing platform.
(303) It should be noted that the landing speed and timing of the plurality of UAVs need to be determined based on the recall mode of the UAVs. For details, reference may be made to the foregoing exemplary embodiments, and it will not be repeated herein.
(304) In summary, the takeoff and landing control method described herein at least may have the following advantages:
(305) In the present disclosure, in response to the received takeoff instruction, the plurality of UAVs take off from the takeoff and landing platform; in response to the received recall instruction, the plurality of UAVs land and are stacked vertically on the takeoff and landing platform along the guide rail. In this way, it is possible to avoid manual UAV take-off site layout, preparation, and landing recovery operations, and reduce manpower and site investment when multiple UAVs perform collaborative operations. It can not only reduce the cost of multiple UAV collaborative operations, but also improve the efficiency of multiple UAV collaborative operations.
(306) The device embodiments described above are only illustrative; the units described as separate components may or may not be physically separate, parts displayed as units may or may not be physical units, they may be located in one place, or may be distributed across multiple network units. Some or all of the modules may be selected according to actual needs to achieve the purpose of the solution of certain embodiments. A person of ordinary skill in the art can understand and implement the methods without any creative efforts.

(307) Reference herein to “one embodiment,” “an embodiment,” or “one or more embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. In addition, it is noted that the expression “in one embodiment” herein does not necessarily refer to the same embodiment.
(308) In the description provided here, a number of specific details are described. However, it is understood that embodiments of the present disclosure may be practiced without these specific details. In some instances, well-known methods, structures, and techniques have not been shown in detail so as not to obscure the understanding of this disclosure.
(309) In the claims, any reference signs placed between parentheses shall not be construed as limiting the claim. The word “comprising” does not exclude the presence of elements or steps not listed in the claim. The word “a” or “an” preceding an element does not exclude the presence of a plurality of such element. The present disclosure may be implemented by means of hardware comprising several different elements and by means of a suitably programmed computer. In a claim enumerating several devices, some of these means may be embodied by the same item of hardware. The use of the words first, second, third, etc. does not indicate any order. These words can be interpreted as names.
(310) Finally, it should be noted that the above exemplary embodiments are only used to illustrate the technical solutions of the present disclosure, but not to limit them. Although the present disclosure has been described in detail with reference to the foregoing exemplary embodiments, a person of ordinary skill in the art will understand as follows: it is still possible to modify the technical solutions disclosed in the foregoing exemplary embodiments, or to equivalently substitute some of the technical features thereof; however, these modifications or substitutions do not cause the essence of the corresponding technical solution to deviate from the principle and scope of the technical solution of each embodiment disclosed in the present disclosure.

Claims

1. A platform for one or more unmanned aerial vehicles (UAVs), comprising: a base; a bracket, comprising a guide rail extended upright from the base; an accommodating portion defined in the bracket and forming a storage space extended along the guide rail to accommodate the one or more UAVs, the accommodating portion comprising an opening for the one or more UAVs to vertically land through or launch from a top of the storage space; a guide member comprising a flow guide frame on a top part of the bracket surrounding the storage space, and the flow guide frame being configured to utilize a downward air flow generated by the one or more UAVs to push the one or more UAVs toward the storage space.
2. The platform according to claim 1, wherein the one or more UAVs comprises a plurality of UAVs; when the plurality of UAVs is stacked vertically on the bracket, the platform has a charging state in which the plurality of UAVs is charged by the platform.
3. The platform according to claim 2, further comprising: one or more first charging modules vertically stacked on the bracket, wherein in the charging state, one of the one or more first charging modules is electrically connected to a second charging module of one of the one or more UAVs for charging.
4. The platform according to claim 3, wherein each of the one or more first charging modules comprises a charging interface and a signal transmission interface; the charging interface is configured to charge the one or more UAVs; and the signal transmission interface is configured to perform data exchange between each of the one or more first charging module and the second charging module.
5. The platform according to claim 2, wherein the platform further has a storage state; and positions of the plurality of UAVs relative to the bracket in the storage state are more compact than positions

of the plurality of UAVs relative to the bracket in the charging state.
6. The platform according to claim 1, wherein the bracket comprises a first bracket and a second bracket spaced apart from each other; and the storage space is formed between the first bracket and the second bracket.
7. The platform according to claim 1, wherein the bracket comprises one bracket, and the storage space is formed on the bracket.
8. The platform according to claim 7, wherein each of the one or more UAVs comprises a mounting through hole; and when the one or more UAVs is vertically landed on the bracket, the bracket is sleeved by the mounting through holes of the one or more UAVs.
9. The platform according to claim 1, wherein the guide rail on the bracket is at least partially embedded in the one or more UAVs to limit movement of the UAVs in the storage space.
10. The platform according to claim 1, further comprising: a guide carrier plate, wherein the guide carrier plate is at least partially disposed in the storage space, and the guide carrier plate is configured to carry the one or more UAVs.
11. The platform according to claim 10, wherein the bracket comprises a lifting module to lift the one or more UAVs, wherein the lifting module is connected to the guide carrier plate, and is configured to slide along the bracket.
12. The platform according to claim 11, further comprising: a control module electrically connected to the lifting module, wherein the control module is configured to: control the lifting module to drive the guide carrier plate to move down to allow the one or more UAVs to vertically stack on the bracket along the guide rail in a landing process of the plurality of the one or more UAVs, or control the lifting module to drive the guide carrier plate to move up to allow the one or more UAVs to take off from the bracket in a takeoff process of the one or more UAVs.
13. The platform according to claim 1, further comprising: the guide member connected to the bracket, wherein the guide member is configured to guide the one or more UAVs into the storage space.
14. The platform according to claim 1, wherein the guide member comprises a plurality of air flow generating devices, wherein the plurality of air flow generating devices is arranged on a top part of the bracket to surround the storage space, and the plurality of air flow generating devices is configured to push the plurality of UAVs toward the storage space.
15. The platform according to claim 1, further comprising: a lifting device; wherein a lifting direction of the lifting device is consistent with an extending direction of the bracket, and the lifting device is configured to transport a target carried by the one or more UAVs when the one or more UAVs is stacked on the bracket.
16. The platform according to claim 1, wherein an auxiliary moving device is disposed on a bottom part of the base, and the auxiliary moving device is configured to assist the platform to move.
17. A docking system, comprising: a platform, comprising: a bracket, comprising a vertical guide rail; a storage space for accommodating one or more unmanned aerial vehicles (UAVs); and a guide member, connected to the bracket, and comprising a flow guide frame, wherein the flow guide frame is arranged on a top part of the bracket to surround the storage space, and the flow guide frame is configured to utilize a downward air flow generated by the one or more UAVs to push the one or more UAVs toward the storage space; and the one or more UAVs, comprising: one of a protruding structure and a recessing structure disposing at a bottom part of the UAV, the other one of the protruding structure and the recessing structure disposing on a top part of the UAV, wherein the protruding structure is configured to at least partially embed in the recessing structure to facilitate the UAV vertical stacking with other UAV of the one or more UAVs, the one or more UAVs is configured to stack vertically on a takeoff and landing platform.